# Design and Evaluation of a Drift-Aware Continuous Learning MLOps Framework for Financial Transaction Fraud Detection

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## Abstract

Fraud models are usually trained once, but in production fraud patterns change (**concept drift**) and true labels arrive weeks late (**label delay**). We build and evaluate a **drift-aware continuous learning MLOps framework** that monitors a deployed model on delayed labels, retrains it on a schedule with an automatically chosen training window, and deploys a new model only if it passes a **promotion gate**. It is tested on three public datasets with label delays of 0 to 60 days.

## Problem Statement

Existing studies test fraud models on **random splits**, assume **immediate labels** and trigger retraining with **drift detectors** on artificial drift. When should a real fraud model be retrained, and how can a bad model be kept out of production?

## Research Objectives

1. Measure how a deployed fraud model decays and whether drift measures detect it.
2. Measure how label delay limits the benefit of retraining.
3. Compare scheduled and drift-triggered retraining fairly.
4. Build an MLOps framework with monitoring, retraining, window selection and a promotion gate, and test it with injected faults.
5. Replicate on three public datasets.

## Literature Review

- **Dal Pozzolo et al.** [1]: fraud labels arrive late (verification latency), which limits how a model can be updated.
- **Menezes and Filho** [2]: fraud models on IEEE-CIS lose up to 40% F1 under natural drift; no retraining tested.
- **Shakil et al.** [3]: drift-triggered retraining for MLOps, tested only on injected drift with immediate labels.

**Gap:** no fair test of retraining policies for fraud under real drift and label delay.

## Data set

| | IEEE-CIS | Sparkov | BAF |
|---|---|---|---|
| Type | Real e-commerce | Simulated cards | Bank accounts |
| Records | 590,540 | 1,801,969 | 1,000,000 |
| Time span | 182 days | 720 days | 8 months |
| Fraud rate | 3.50% | 0.53% | 1.10% |
| Features | 431 | 16 | 29 |

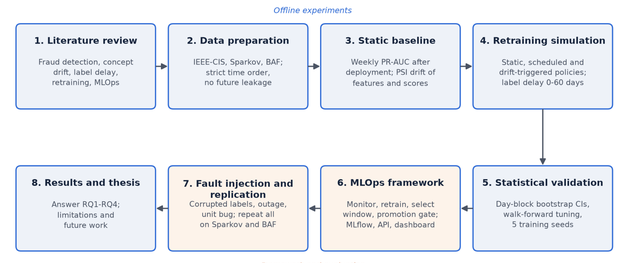
### Preprocessing

- **IEEE-CIS:** transaction and identity tables joined on TransactionID; text columns as categories; missing values kept.
- **Sparkov:** personal fields removed; cut at 21 Dec 2020 (fraud stops); age, hour, distance and card-history features from *earlier* transactions only.
- **BAF:** -1 codes set to missing; month not used as a feature.
- **All:** sorted by time, never shuffled; no oversampling (fraud class weighted).

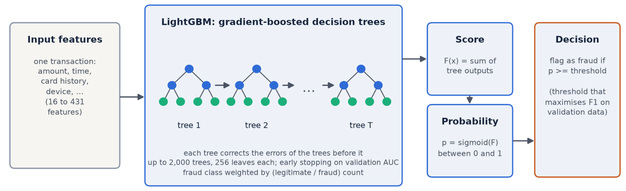



Preprocessing pipeline from raw files to the time-ordered table used by every experiment

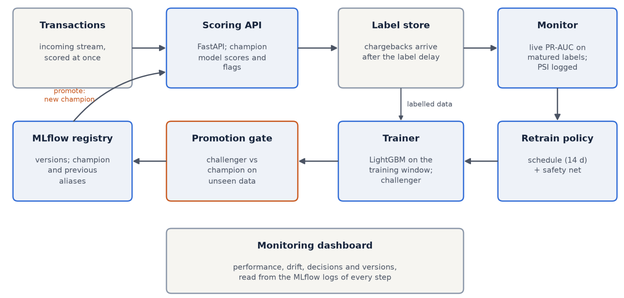
## Methodology




Workflow of the study

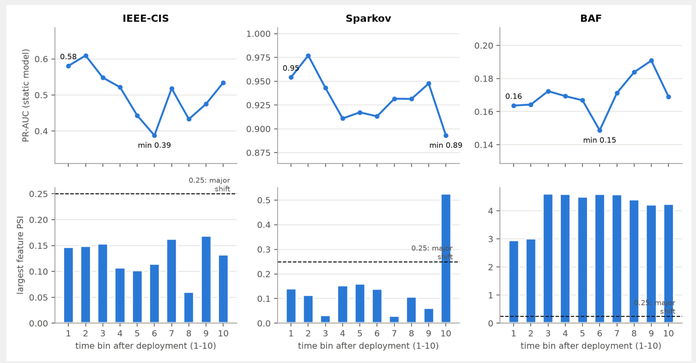



Model architecture: LightGBM gradient-boosted decision trees

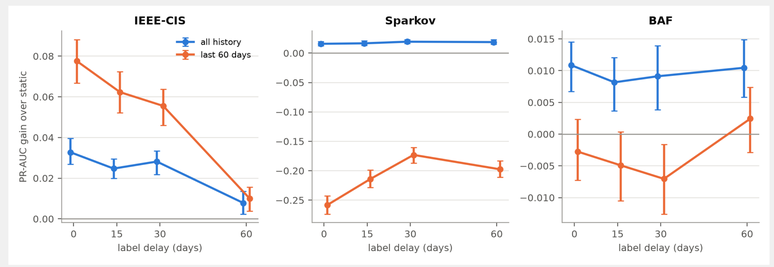



Proposed framework: each week it monitors, decides, retrains, gates and serves

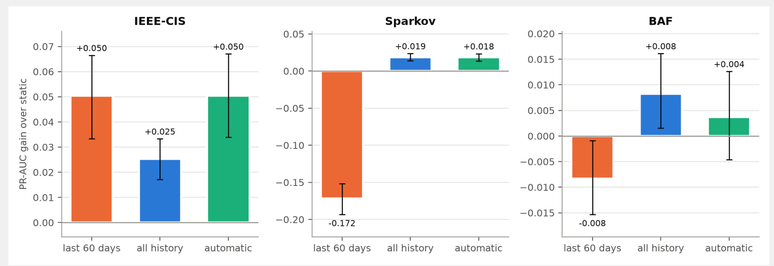
## Result Analysis



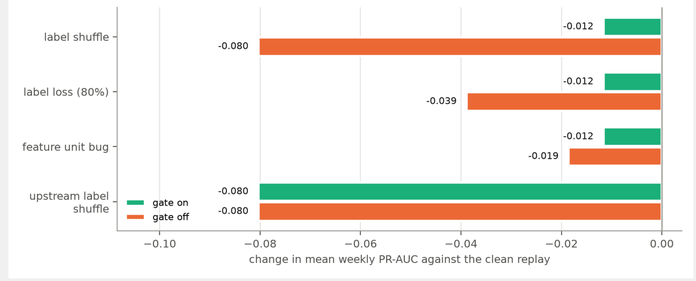

**Drift measures do not track performance:** IEEE-CIS decays while drift stays low; Sparkov and BAF drift without decaying




**Label delay erodes retraining:** on IEEE-CIS the gain falls from +0.078 to +0.010 PR-AUC




**Automatic window selection** matches the best fixed window without hindsight




**The promotion gate** blocked every model trained on corrupted data

## Conclusion & Future Work

- **Drift indicators can miss real degradation under delayed labels**, while scheduled retraining was more reliable. Here, *drift-aware* means tracking drift through delayed-label performance, not label-free alarms.
- With delayed labels, the schedule matched or beat the drift trigger even at an **equal retraining budget** (IEEE-CIS, BAF): not only from retraining more often.
- **Window selection** and a **promotion gate** make retraining adaptive and safe; results replicated on **two more timestamped datasets** (Sparkov, BAF).
- **API load test** (supervisor feedback): after a fix to the scoring path, about **1,300 requests/s**, p95 latency 82 ms at 100 clients, identical scores. **Next:** equal-budget comparison on Sparkov.
- **Later:** faster investigator feedback, more model families, explanation monitoring and a live deployment.

## References

[1] A. Dal Pozzolo et al., "Credit card fraud detection: A realistic modeling and a novel learning strategy," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, 2018.

[2] R. S. Menezes and R. H. Filho, "Semantic and structural drift in financial knowledge graphs: A robustness analysis of GNN-based fraud detectors," in *Proc. IEEE ICKG*, 2025.

[3] M. N. P. Shakil et al., "Feature drift-guided adaptive ML retraining: An MLOps approach for big data analytics," in *Proc. IEEE BigData*, 2025.


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